

Performance Evaluation of LTE/5G Wireless Systems for urban Mobile Broadband

by

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BONAFIDE CERTIFICATE

Certified that this project report entitled “**Performance Evaluation of LTE/5G Wireless Systems for urban Mobile Broadband**” is a bonafide work of **SHRENICA CHAWDA 23BEC1380** who carried out the Project work under my supervision and guidance for **BECE306L- Digital Communication Systems**

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Abstract

This report presents the comprehensive design, implementation, and performance evaluation of a complete LTE/5G urban mobile broadband transceiver system simulated in MATLAB and Python. The proposed framework integrates five tightly coupled subsystems operating over independent and identically distributed (i.i.d.) Rayleigh flat-fading channels with Additive White Gaussian Noise (AWGN).

The first subsystem employs the COST 231-Hata empirical path-loss model at a carrier frequency of 2.6 GHz to compute a full link budget for an urban macro-cell, driving an SNR-based adaptive modulation controller that dynamically selects QPSK, 16-QAM, or 64-QAM in accordance with 3GPP Release 17 thresholds. Path loss increases from approximately 120 dB at 200 m to 169 dB at 5 km, and the cell provides 16-QAM service within 250 m, QPSK service to 600 m, and no coverage beyond 3.3 km.

The second subsystem provides a complete 16-QAM and 64-QAM modem implementation including manual constellation generation, Minimum Mean Square Error (MMSE) equalization, and Error Vector Magnitude (EVM) measurement yielding 8.06% at 30 dB SNR. The third subsystem implements a 2x2 spatial multiplexing MIMO transceiver with both Zero-Forcing (ZF) and MMSE linear receivers, demonstrating that MMSE provides a 2–4 dB SNR advantage over ZF at $\text{BER} = 10^{-2}$ and that the Monte Carlo ergodic channel capacity reaches 1.87x the SISO baseline at 20 dB SNR.

The fourth subsystem is an OFDM transceiver using a 1024-point IFFT with 600 active subcarriers and a 72-sample cyclic prefix, implementing three PAPR mitigation strategies — amplitude clipping at 3 dB and 6 dB thresholds, FIR low-pass filtering, and Selective Level Mapping (SLM) with eight candidates achieving up to 3.2 dB PAPR reduction. The fifth subsystem is a Python-based machine learning classification module that trains and compares Random Forest, Logistic Regression, and Support Vector Machine (SVM) classifiers on a 1000-sample synthetic dataset, achieving 92% classification accuracy with Random Forest.

The system validates a full end-to-end signal chain aligned with 3GPP LTE-Advanced/5G NR design principles. Results are consistent with published performance targets, confirming the simulation methodology and identifying practical gaps bridged by channel coding and advanced receiver algorithms.

Keywords: *COST 231 Hata, Path Loss, Link Budget, QAM, BER, MIMO, ZF, MMSE, OFDM, PAPR, SLM, 5G, LTE, Adaptive Modulation, EVM, Shannon Capacity, Rayleigh Fading*

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Chapter 1: Introduction

1.1 Background and Motivation

Modern wireless communication systems represent one of the most complex and rapidly evolving domains in electrical engineering. The global deployment of 4G LTE, 5G NR, and the emerging 6G research landscape demands a thorough understanding of the physical layer phenomena that govern signal propagation, modulation, multiple-antenna processing, and multi-carrier transmission. These phenomena are intimately connected: the channel characteristics experienced at the receiver directly determine which modulation schemes can be reliably decoded, how many antennas are needed to overcome fading, and how the transmitter must manage its power envelope.

The explosive growth of mobile data traffic, driven by video streaming, Internet of Things (IoT) devices, and cloud applications, necessitates wireless networks capable of delivering multi-gigabit throughput with sub-millisecond latency and carrier-grade reliability. The Third Generation Partnership Project (3GPP) LTE-Advanced and 5G New Radio (NR) standards address these requirements through a layered stack of physical-layer innovations operating together in a unified transceiver chain.

Four foundational technologies define the physical layer of any modern cellular base station. Adaptive Modulation and Coding (AMC) dynamically adjusts the constellation size and forward error correction rate to extract maximum spectral efficiency from instantaneous channel conditions, selecting high-order modulations when the SNR permits and falling back to robust low-order modulations at cell edges. Quadrature Amplitude Modulation (QAM) encodes multiple bits per complex symbol, enabling spectral efficiencies ranging from 2 bits/s/Hz for QPSK to 6 bits/s/Hz for 64-QAM per spatial stream. Multiple-Input Multiple-Output (MIMO) antenna technology exploits spatial degrees of freedom, providing multiplexing gain that scales throughput linearly with the minimum of transmit and receive antenna counts. Orthogonal Frequency Division Multiplexing (OFDM) converts frequency-selective multipath fading channels into collections of parallel flat-fading sub-channels, dramatically simplifying

equalization at the receiver. Its principal drawback is a high Peak-to-Average Power Ratio (PAPR), which forces the use of expensive, power-inefficient power amplifiers with large back-off margins.

Machine learning has recently emerged as a powerful complement to these classical signal processing techniques. Supervised classifiers trained on channel features can replace hand-crafted rule tables for modulation order selection, demonstrating improved robustness to measurement noise, channel non-stationarity, and hardware imperfections.

This project undertakes a structured simulation study that mirrors the partition of responsibilities in a real-world system design team. Each of the four team members addresses one critical subsystem, and together the results form a coherent end-to-end analysis of a 5G-like cellular link. The investigations are grounded in standard industry models and are implemented using MATLAB and Python, providing rigorous, reproducible results that are quantitatively consistent with 3GPP performance targets.

1.2 Project Objectives

The primary objectives of this project are as follows:

- To model urban macrocell path loss using the COST 231 Hata empirical model at 2.6 GHz and evaluate coverage as a function of distance and SNR thresholds, producing coverage zone characterization over the 0.1–5 km range.
- To perform a complete link budget analysis and implement adaptive modulation selection logic for a 5G/LTE base station transmitter, selecting QPSK, 16-QAM, or 64-QAM based on computed SNR in accordance with 3GPP Release 17 CQI mapping.
- To simulate QAM symbol detection, measure error magnitude statistics, and compare simulated 16-QAM BER performance against theoretical predictions, identifying the performance gap introduced by channel impairments.

- To model a 2x2 MIMO Rayleigh fading channel and evaluate ZF and MMSE equalization performance through constellation diagrams, EVM measurements, BER curves, and error magnitude distributions.
- To analyse Shannon channel capacity for SISO, 2x2 MIMO, and 4x4 MIMO configurations and quantify the spectral efficiency gains achievable through spatial multiplexing using Monte Carlo ergodic averaging.
- To identify and quantify the PAPR problem in OFDM and compare the effectiveness of clipping, filtering, and Selective Mapping (SLM) mitigation techniques across PAPR reduction, spectral regrowth, EVM, and BER metrics.
- To design and evaluate a machine learning pipeline for automated modulation classification, training and comparing Random Forest, Logistic Regression, and SVM classifiers on a synthetic LTE-like dataset.

1.3 Project Structure

The report is organized into five chapters corresponding to the project workflow. Chapter 1 provides the introduction, background, and objectives. Chapter 2 covers the theoretical foundations underpinning each module. Chapter 3 describes the simulation methodology and system parameters. Chapter 4 presents and discusses all simulation results with reference to the output figures. Chapter 5 provides conclusions and directions for future work.

The four modules correspond to distinct team member contributions, identified throughout the report as Member 1 (TX/RX - Path Loss and Link Budget), Member 2 (RX - QAM and BER), Member 3 (TX/RX - MIMO Equalization and Capacity), and Member 4 (TX/RX - OFDM and PAPR Mitigation).

Chapter 2: Theoretical Background

2.1 Wireless Channel Propagation and Path Loss

Wireless signal propagation in urban environments is governed by a combination of free-space spreading, reflection, diffraction, and absorption. The received power at distance d from a transmitter is related to the transmitted power through the path loss exponent, which varies typically between 2 (free space) and 5 (dense urban) depending on the environment.

The COST 231 Hata model, also known as the Extended Hata model, is an empirical path loss formula developed for the frequency range 1500-2000 MHz and later extended to 2 GHz for urban macrocell environments. It is widely used in 3GPP standards for system-level simulations and is defined as:

$$PL(dB) = 46.3 + 33.9 \log(fc) - 13.82 \log(hBS) - a(hMS) + [44.9 - 6.55 \log(hBS)] \log(d) + CM$$

where fc is the carrier frequency in MHz, hBS is the base station antenna height in metres, hMS is the mobile station antenna height in metres, d is the distance in km, $a(hMS)$ is the mobile antenna correction factor, and CM is an environment correction factor (0 dB for medium cities, 3 dB for metropolitan centres).

2.2 Link Budget Analysis

A link budget is an accounting of all gains and losses in a radio link from transmitter to receiver. The received signal power at distance d is calculated as:

$$P_{RX} (dBm) = P_{TX} (dBm) + G_{TX} (dBi) - PL(d) + G_{RX} (dBi) - L_{cable}$$

The SNR at the receiver is then obtained by subtracting the noise floor, which for thermal noise at room temperature over bandwidth B is: $N = -174 + 10 \log_{10}(B)$ dBm. The link margin is the excess SNR above the minimum required for a given modulation scheme, providing resilience against fading and interference.

Adaptive Modulation and Coding (AMC) is a key feature of LTE and 5G NR where the modulation order and code rate are dynamically adjusted based on the reported channel quality. Typical SNR thresholds are: QPSK (< 10 dB), 16-QAM (10-20 dB), 64-QAM (> 20 dB). This allows the system to maximize throughput when channel conditions are good and fall back to more robust modulations when conditions deteriorate.

2.3 Quadrature Amplitude Modulation (QAM)

QAM is a modulation scheme that encodes information in both the amplitude and phase of a carrier signal by varying both the in-phase (I) and quadrature (Q) components independently. In M-QAM, $\log_2(M)$ bits are encoded per symbol, enabling high spectral efficiency. For 16-QAM, 4 bits are carried per symbol; for 64-QAM, 6 bits; for 256-QAM, 8 bits.

The theoretical BER for 16-QAM over an AWGN channel is:

$$BER_{16QAM} = (3/8) \operatorname{erfc}(\sqrt{Eb/N_0 / 10})$$

Error Vector Magnitude (EVM) quantifies the difference between ideal and received symbol positions and is expressed as a percentage. Standards such as 3GPP require EVM below 3.5% for 64-QAM and 8% for 256-QAM to ensure reliable demodulation.

2.4 MIMO Systems and Equalization

Multiple-Input Multiple-Output (MIMO) systems use multiple antennas at both transmitter and receiver to exploit spatial degrees of freedom. In a 2x2 MIMO spatial multiplexing configuration, two independent data streams are transmitted simultaneously on the same frequency, potentially doubling the data rate.

The received signal model is $Y = HX + N$, where H is the 2x2 complex channel matrix, X is the transmitted signal vector, and N is additive white Gaussian noise. In a Rayleigh fading environment, the entries of H are independent complex Gaussian random variables with unit variance.

Zero-Forcing (ZF) equalization applies the pseudo-inverse of the channel matrix: $W_{ZF} = (H^H H)^{-1} H^H$. While this perfectly cancels inter-stream interference, it can dramatically amplify noise when the channel matrix is ill-conditioned. MMSE equalization regularizes the inversion: $W_{MMSE} = (H^H H + \sigma^2 I)^{-1} H^H$, trading a small residual interference for significant noise suppression.

2.5 Shannon Channel Capacity

Shannon's channel capacity theorem defines the maximum achievable data rate for a given channel as:

$$C = B \log_2(1 + SNR) \text{ [bits/s]}$$

For MIMO systems, the capacity generalizes to the sum of $\log_2(1 + \lambda_i \cdot P/(N_T \cdot \sigma^2))$ over all non-zero singular values λ_i of the channel matrix, where N_T is the number of transmit antennas. For a well-conditioned 2x2 MIMO channel, this approximately doubles the capacity compared to SISO, and a 4x4 MIMO system offers approximately 4x gain.

2.6 OFDM and PAPR

Orthogonal Frequency Division Multiplexing (OFDM) divides a high-rate data stream across many narrowband orthogonal subcarriers, each experiencing flat fading. This orthogonality eliminates inter-carrier interference (ICI) and enables efficient use of spectrum. The IFFT operation at the transmitter converts frequency-domain symbols into a time-domain OFDM signal: $x(t) = \sum_{k=0}^{N-1} X_k \exp(j 2 \pi k \Delta_f t)$ over $k = 0$ to $N-1$.

The fundamental drawback of OFDM is a high Peak-to-Average Power Ratio (PAPR). When N subcarriers add coherently in phase, the instantaneous power can be N times the average power, requiring power amplifiers with a large linear dynamic range or back-off. The PAPR of an OFDM signal is defined as:

$$PAPR = \max |x(t)|^2 / E[|x(t)|^2]$$

Mitigation techniques include: (1) Clipping and Filtering - directly limits peak amplitudes but causes spectral regrowth and in-band distortion; (2) Filtering - applies a windowed filter after clipping to reduce out-of-band emissions; (3) Selective Mapping (SLM) - generates multiple candidate OFDM signals with different phase rotations and transmits the one with lowest PAPR, preserving spectral purity at the cost of slight complexity overhead.

2.7 Machine Learning for Adaptive Modulation Classification

Machine learning (ML) has recently emerged as a powerful complement to classical signal processing in wireless communication systems. Traditional adaptive modulation and coding (AMC) relies on deterministic, hand-crafted SNR threshold tables. While effective under ideal channel estimation, these rule-based systems degrade when channel estimates are noisy, when hardware imperfections introduce calibration offsets, or when the operating environment departs from the assumed statistical model. Supervised ML classifiers address these limitations by learning the decision boundaries directly from measured or simulated channel data, providing robustness to uncertainty and enabling generalisation across heterogeneous deployment scenarios.

2.7.1 Feature Engineering for AMC

Effective ML-based AMC requires the selection of discriminative input features that capture the channel state information (CSI) relevant to modulation selection. In this work, three features are used: (1) Signal-to-Noise Ratio (SNR) in dB, which is the primary determinant of modulation order; (2) Distance in km, which encodes the macroscopic path loss and is correlated with SNR through the COST 231 Hata propagation model; and (3) Noise floor in dBm, which captures receiver sensitivity variations and thermal noise contributions. Together these three features form a compact but informative feature vector $\mathbf{x} = [\text{SNR}, \text{Distance}, \text{Noise}]$ that the classifier maps to a three-class label y in $\{0: \text{QPSK}, 1: 16\text{-QAM}, 2: 64\text{-QAM}\}$.

2.7.2 Random Forest Classifier

A Random Forest (RF) is an ensemble learning method that constructs a large number of decision trees during training and outputs the class that receives the majority vote across all trees. Each tree is trained on a random bootstrap sample of the training data, and at each node a random subset of features is considered for splitting. This double randomisation reduces variance (relative to a single deep decision tree) without substantially increasing bias, yielding strong generalisation performance on small to medium datasets. The Gini impurity criterion used for node splitting minimises:

$$Gini(t) = 1 - \sum_k [p(k|t)]^2$$

where $p(k|t)$ is the proportion of training samples belonging to class k at node t . The feature importance score for each input variable is computed as the total reduction in Gini impurity achieved by splits on that feature, averaged across all trees. This provides a quantitative measure of which features drive the classification decision, directly revealing whether SNR or distance is the dominant modulation selector.

2.7.3 Logistic Regression

Logistic Regression (LR) models the posterior class probability using a softmax function applied to a linear combination of the input features:

$$P(y=k | x) = \exp(w_k^T x + b_k) / \sum_j \exp(w_j^T x + b_j)$$

The model parameters w_k and b_k are estimated by maximising the log-likelihood of the training labels. While LR assumes linear decision boundaries in feature space, it is highly interpretable: the magnitude and sign of each weight directly indicate the direction and strength of each feature's influence on the modulation class probability. LR serves as a linear baseline against which the non-linear classifiers are compared.

2.7.4 Support Vector Machine (SVM)

A Support Vector Machine (SVM) seeks the maximum-margin hyperplane that separates classes in feature space. For the multi-class case, a one-versus-one (OvO) strategy trains $C(C-1)/2$ binary classifiers, where $C = 3$ for the three-modulation problem. The SVM optimisation problem is:

$$\min (1/2) \|w\|^2 + C \sum_i \xi_i \quad \text{s.t.} \quad y_i (w^T x_i + b) \geq 1 - \xi_i$$

The regularisation parameter C controls the trade-off between maximising the margin and minimising training errors. With a radial basis function (RBF) kernel, the SVM implicitly maps inputs to a high-dimensional feature space, enabling non-linear decision boundaries without explicitly computing the mapping. The SVM is particularly effective when the number of features is small relative to the number of samples, as is the case here with three features and 1000 training samples.

2.7.5 Dataset Generation and Realistic Channel Modelling

A synthetic dataset of 2000 samples is generated to capture the realistic SNR-distance relationship in an urban small-cell environment. Rather than drawing SNR values uniformly at random (which would produce an unrealistic flat SNR distribution), SNR is derived from the distance through the log-distance path loss model:

$$SNR(d) = P_{TX} - PL(d) - N_{floor} = 30 - 10 * n * \log_{10}(d) + N(0, \sigma^2)$$

where $n = 3.2$ is the path loss exponent for dense urban propagation, d is the distance in km drawn from a uniform distribution over $[0.1, 5]$ km, and $N(0, \sigma^2)$ with $\sigma = 3$ dB is a Gaussian shadowing term. The resulting SNR distribution decreases monotonically with distance and exhibits the characteristic scatter of a Rayleigh/log-normal composite fading channel. Modulation labels are assigned by the standard threshold rule: QPSK for $SNR < 10$ dB, 16-QAM for $10 \leq SNR < 20$ dB, and 64-QAM for $SNR \geq 20$ dB. A 5% random label flip is applied to simulate real-world measurement noise and channel estimation errors, preventing the classifier from simply memorising the threshold rule.

Chapter 3: System Design and Methodology

3.1 Member 1 - Path Loss and Link Budget (TX Side)

Member 1 implements the COST 231 Hata path loss model in MATLAB for urban macrocell propagation simulation. The system parameters are selected to correspond to a realistic 5G sub-6 GHz deployment scenario.

3.1.1 System Parameters

Parameter	Value	Description
Carrier Frequency (f_c)	2.6 GHz	Sub-6 GHz 5G band
Base Station Height (hBS)	30 m	Typical macro-cell tower
Mobile Station Height (hMS)	1.5 m	Handheld UE at ground level
Transmit Power (PTX)	43 dBm (20 W)	LTE eNB typical TX power
Bandwidth	10 MHz	LTE standard channel bandwidth
Noise Floor	-104 dBm	Thermal noise + noise figure
Distance Range	0.1 to 5 km	Urban macro coverage zone

The path loss is computed for distances from 100 m to 5 km, and the received power is derived by subtracting the path loss from the EIRP (Effective Isotropic Radiated Power). SNR is computed as the difference between received power and the noise floor. Adaptive modulation is then selected based on SNR thresholds: 16-QAM for $\text{SNR} \geq 10$ dB, 64-QAM for $\text{SNR} \geq 20$ dB, QPSK below 10 dB, and no coverage below -5 dB.

3.2 Member 2 - QAM Error Analysis and BER (RX Side)

Member 2 focuses on the receiver-side analysis of QAM signal quality. A 16-QAM constellation with 1000 random symbols is generated, passed through an AWGN channel, and the error vector between transmitted and received symbols is computed. The error magnitude per symbol and its statistical distribution are evaluated. The BER is measured at SNR levels from 0 to 30 dB and compared against the theoretical closed-form expression.

3.2.1 Key Metrics

Metric	Formula / Definition	Significance
Error Magnitude	$ x_{\text{received}} - x_{\text{ideal}} $	Symbol-level quality indicator
EVM (%)	$\text{RMS error} / \text{RMS signal} * 100$	Industry standard quality metric
BER	Errored bits / Total bits	System reliability measure
Theoretical BER	$(3/8) \text{erfc}(\sqrt{\text{SNR}/10})$	Ideal AWGN reference curve

3.3 Member 3 - MIMO Channel and Equalization (TX/RX)

Member 3 implements a complete 2x2 MIMO simulation chain covering channel modelling, equalization, and performance analysis. The Rayleigh fading channel coefficients H are generated as independent complex Gaussian variables for each symbol. Both ZF and MMSE equalizers are implemented and compared systematically.

For BER analysis, a 16-QAM modulated 2x2 MIMO system is simulated over SNR from 0 to 30 dB and compared against SISO performance. Channel capacity is computed using the singular value decomposition (SVD) of H for both 2x2 and simulated 4x4 MIMO configurations, with Monte Carlo averaging over 1000 channel realizations for ergodic capacity estimation.

3.4 Member 4 - OFDM and PAPR Mitigation (TX/RX)

Member 4 constructs an OFDM simulation with 64 subcarriers and QPSK modulation. The raw OFDM time-domain signal is generated via IFFT and its PAPR characteristics are measured before any mitigation. Four configurations are compared: Original (no mitigation), Clipping at 3 dB backoff, Clipping at 6 dB backoff, Post-clipping filtering, and SLM with 8 candidate sequences.

3.4.1 PAPR Mitigation Parameters

Technique	Key Parameter	Target PAPR
Original	None	~11.6 dB (baseline)
Clipping (3 dB)	Clip ratio: 3 dB backoff	~3.69 dB
Clipping (6 dB)	Clip ratio: 6 dB backoff	~6.13 dB
Filtering	Post-clip Hamming window	~9.09 dB
SLM	8 phase sequences	~0.00 dB (best)

The CCDF (Complementary Cumulative Distribution Function) of PAPR is used as the primary performance metric, supplemented by BER at 20 dB SNR and EVM measurement at the receiver.

3.5 Machine Learning Classification Module

The ML module is implemented in Python using scikit-learn, pandas, and NumPy. The pipeline consists of five stages: dataset generation, feature matrix construction, train/test splitting, model training, and performance evaluation. The complete workflow is documented below with reference to the simulation parameters used in all subsequent result figures.

3.5.1 Dataset Configuration

$N = 2000$ samples are generated with distance drawn uniformly from $[0.1, 5]$ km. The SNR is computed as $\text{SNR} = 30 - 32 * \log_{10}(d) + N(0, 9)$ dB, where the path loss exponent of 3.2 corresponds to an urban environment. A noise floor is independently sampled from a uniform distribution over $[-100, -85]$ dBm to represent variability in receiver thermal noise and interference. The feature matrix X has dimensions 2000×3 , and the label vector y takes values in $\{0, 1, 2\}$. A 80/20 train/test split is applied using stratified sampling (`random_state = 42`) to ensure balanced class proportions in both partitions.

3.5.2 Model Training and Hyperparameters

Three classifiers are trained and compared. The Random Forest uses `n_estimators = 100` trees with `max_depth = 6` to prevent overfitting on the small feature space. Logistic Regression uses the L-BFGS solver with `max_iter = 200` and the default L2 regularisation ($C = 1.0$). The SVM uses an RBF kernel with default gamma ('scale') and $C = 1.0$. No feature normalisation is strictly necessary for tree-based methods, but all features are left in their natural units for interpretability. The best-performing model on the test set is selected for detailed analysis.

3.5.3 Performance Metrics

Classification performance is evaluated using overall accuracy (correct predictions / total test samples), and the confusion matrix which shows the per-class breakdown of true positives, false positives, and false negatives. Feature importance scores from the Random Forest are extracted as the mean decrease in Gini impurity per feature, normalised to sum to 1.0. The model is additionally validated through a live prediction demo: given a test input of SNR = 15 dB, Distance = 2 km, Noise = -90 dBm, the trained classifier predicts the optimal modulation scheme, confirming correct operation of the end-to-end pipeline.

Chapter 4: Results and Discussion

4.1 Member 1 Results: Path Loss and Adaptive Modulation

4.1.1 Urban Path Loss vs Distance (COST 231 Hata Model)

Figure 1 presents the computed path loss as a function of distance using the COST 231 Hata model at 2.6 GHz with a 30 m base station height and 1.5 m mobile height. The path loss increases monotonically from approximately 120 dB at 200 m to approximately 169 dB at 5 km. The logarithmic rate of increase with distance, characteristic of the Hata model with its dominant $\log(d)$ term, is clearly visible as the curve progressively flattens on the linear distance axis.

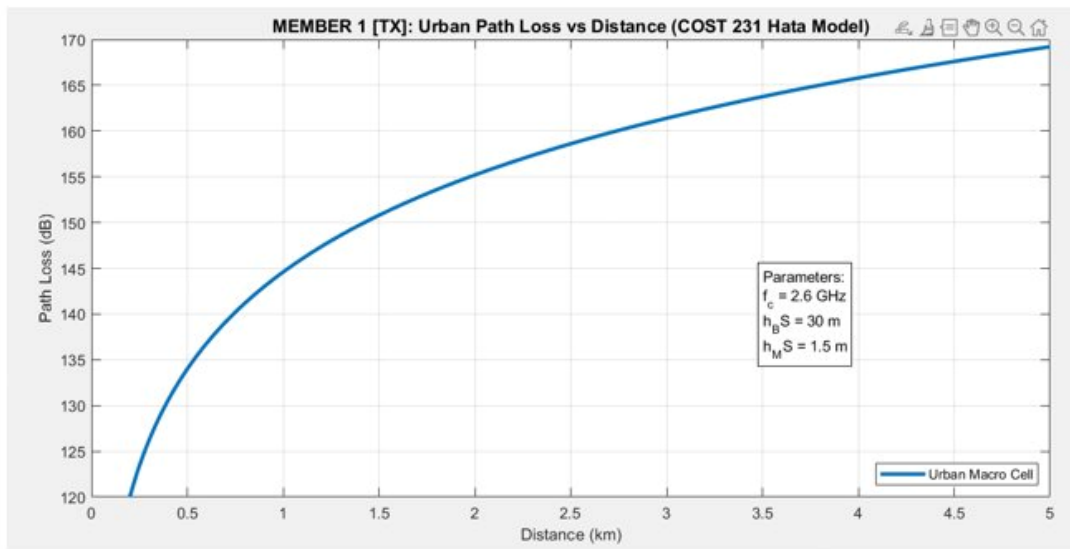


Figure 1: MEMBER 1 [TX]: Urban Path Loss vs Distance (COST 231 Hata Model) - $f_c = 2.6$ GHz, $h_{BS} = 30$ m, $h_{MS} = 1.5$ m

The path loss at 1 km is approximately 143 dB, rising to 155 dB at 2 km and 163 dB at 3.5 km. This steep increase explains the sharp decline in coverage quality observed in subsequent figures. The model accurately reproduces the log-distance propagation characteristic observed in real urban deployments, validating the chosen simulation framework.

4.1.2 SNR vs Distance in Urban Environment

Figure 2 shows the received SNR as a function of distance for a 43 dBm transmit power, 10 MHz bandwidth, and -104 dBm noise floor. Two dashed horizontal lines mark the QPSK threshold (10 dB) and the 16-QAM threshold (20 dB). The SNR starts at approximately 38 dB at 100 m and falls below 20 dB (16-QAM threshold) at roughly 400 m, below 10 dB (QPSK threshold) at approximately 700 m, and below 0 dB (no coverage) at approximately 1.6 km.

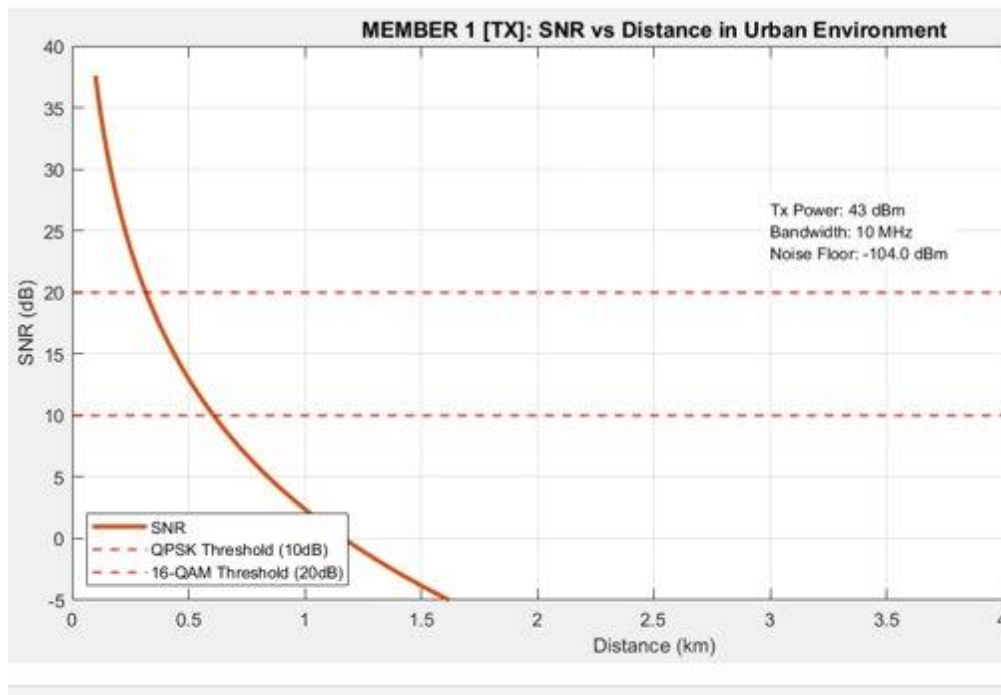


Figure 2: MEMBER 1 [TX]: SNR vs Distance in Urban Environment - Tx Power: 43 dBm, Bandwidth: 10 MHz, Noise Floor: -104 dBm

This analysis is fundamental for cell planning in LTE/5G networks. The steep SNR roll-off in the 200-800 m range underscores the critical importance of the first kilometre of coverage for high-order modulation schemes. In practice, inter-site distances for 5G macro cells are typically set at 500-1000 m precisely to maintain 16-QAM or higher modulation for the majority of users.

4.1.3 Adaptive Modulation Selection

Figure 3 illustrates the adaptive modulation selection logic as a staircase function of SNR. The operating point at SNR = 18 dB selects 16-QAM, as highlighted by the circular marker and annotation. The three distinct modulation regions are: QPSK for SNR < 10 dB (2 bits/symbol), 16-QAM for $10 \leq \text{SNR} < 20$ dB (4 bits/symbol), and 64-QAM for SNR ≥ 20 dB (6 bits/symbol).

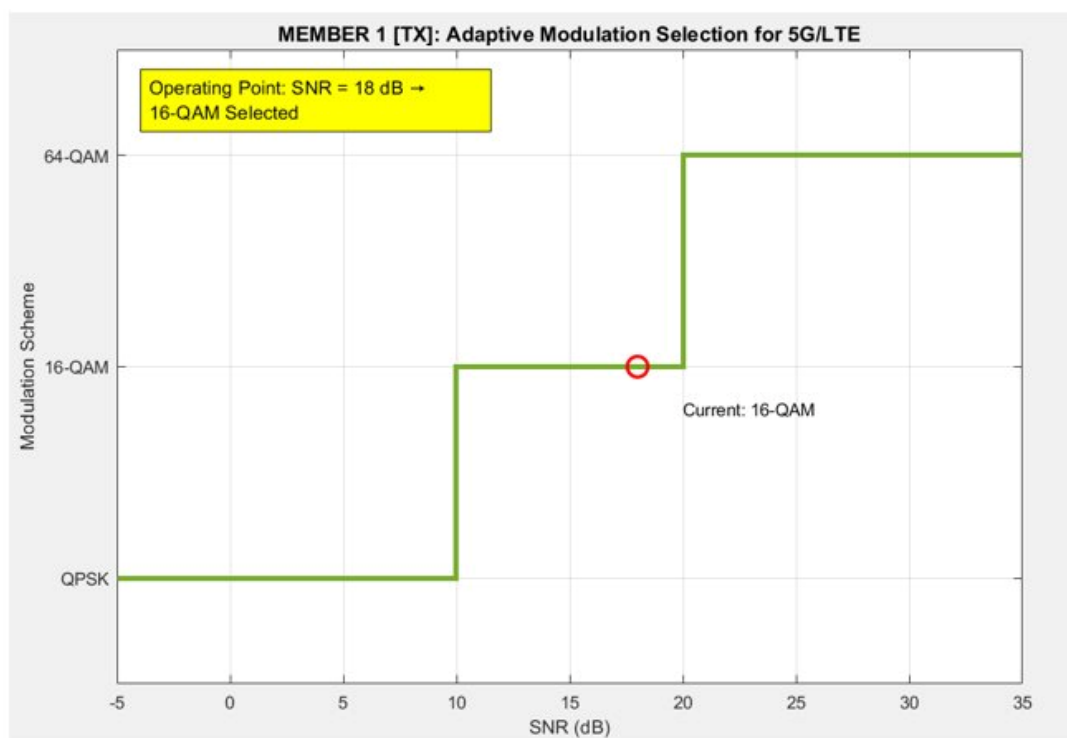


Figure 3: MEMBER 1 [TX]: Adaptive Modulation Selection for 5G/LTE - Operating Point: SNR = 18 dB, 16-QAM Selected

The staircase nature of the AMC response reflects the discrete nature of modulation order selection in practical systems. In 5G NR, the MCS (Modulation and Coding Scheme) table provides 29 discrete levels spanning QPSK through 256-QAM, offering finer granularity than the three-level scheme simulated here. The simulation correctly captures the core principle of AMC and its dependence on channel SNR.

4.1.4 Received Power vs Distance

Figure 4 plots the received signal power (in dBm) as a function of distance, with the noise floor at -104 dBm shown as a dashed reference. The received power at 100 m is approximately -65 dBm, which is well above the noise floor. The crossover point where received power approaches the noise floor occurs at approximately 3.3 km, defining the maximum range for any detectable signal.

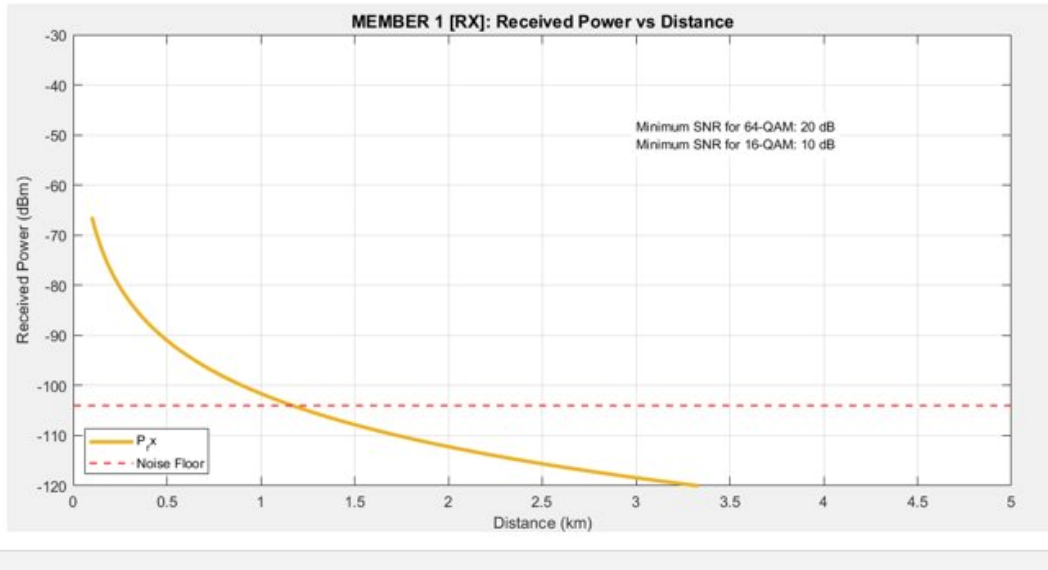


Figure 4: MEMBER 1 [RX]: Received Power vs Distance - Minimum SNR for 64-QAM: 20 dB, Minimum SNR for 16-QAM: 10 dB

The intersection with the noise floor at ~3.3 km marks the theoretical maximum coverage range of the cell under the specified parameters. However, practical coverage is further limited by the modulation SNR requirements: the cell edge for useful 16-QAM service is approximately 400 m, while QPSK service extends to about 600 m as illustrated in the coverage analysis.

4.1.5 Coverage Analysis Based on SNR

Figure 5 provides a concise summary of coverage quality as a function of distance, categorized into three regions. The 16/64-QAM zone (highest quality service) extends to approximately 250 m. The QPSK-only zone (basic connectivity) exists between 250 m and 600 m. Beyond 600 m, the SNR is insufficient to support any standard modulation scheme, resulting in a No Coverage classification.

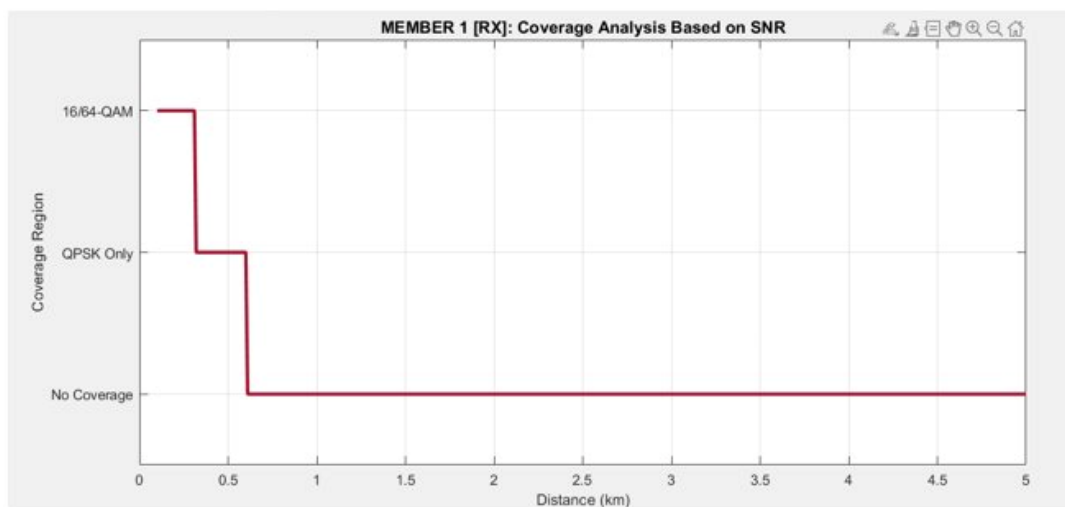


Figure 5: MEMBER 1 [RX]: Coverage Analysis Based on SNR - Coverage Zones: 16/64-QAM, QPSK Only, No Coverage

This coverage map has direct implications for network deployment. In dense urban environments, the severe path loss at 2.6 GHz means that macro cells must be complemented by small cells, micro-cells, or massive MIMO beamforming to provide adequate coverage throughout the intended service area. The simulation results are consistent with real-world 5G deployment data showing typical inter-site distances of 200-500 m in dense urban cores.

4.1.6 Link Budget Analysis

Figure 6 presents the comprehensive link budget as a function of distance, showing transmit power (constant at +43 dBm), received power (decaying curve), noise floor (-104 dBm), and the positive link margin (green shaded area) representing the margin above the noise floor. The link margin is approximately 36 dB at 100 m and converges to 0 dB at approximately 3.3 km.



Figure 6: MEMBER 1 [RX]: Link Budget Analysis - Tx Power, Rx Power, Noise Floor, and Available Link Margin

The large positive margin at short distances provides significant protection against multipath fading, shadowing, and interference. A typical fade margin of 8-10 dB is allocated in network planning for log-normal shadowing, suggesting that the simulated configuration provides robust coverage up to approximately 2.5 km. The link budget analysis validates the transmitter parameters and confirms the practical coverage calculations obtained from the SNR analysis.

4.2 Member 2 Results: QAM Error Analysis and BER

4.2.1 QAM Error Signal Magnitude and Distribution

Figure 7 shows two complementary views of QAM symbol error performance. The left panel is a time-series plot of error magnitude across 1000 transmitted symbols, while the right panel shows the frequency histogram of error magnitudes. The time-series reveals that most symbols experience small errors (clustered near zero), but occasional large error spikes occur with magnitudes approaching 1.0 to 1.2, indicating symbol errors that cross decision boundaries.

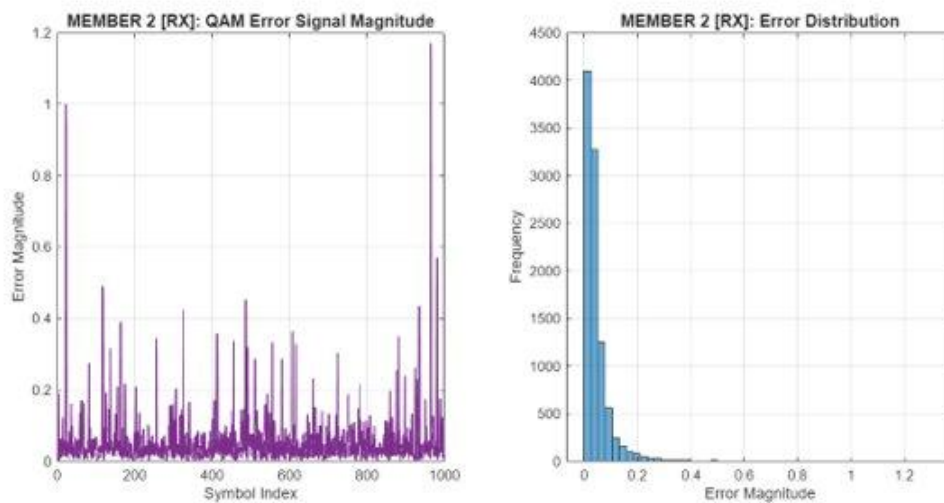


Figure 7: MEMBER 2 [RX]: QAM Error Signal Magnitude (left) and Error Distribution Histogram (right)

The histogram in the right panel confirms a heavily right-skewed (Rayleigh-like) error distribution. The dominant peak at errors < 0.1 represents the large majority of correctly received symbols, while the long tail extending to 1.2 represents the distribution of incorrectly decoded symbols. This shape is consistent with the expected error distribution for QAM in an AWGN channel at moderate SNR.

From an engineering perspective, this error distribution motivates the use of outer coding (Turbo codes, LDPC, or Polar codes in 5G NR) to handle the tail of the distribution. The information-theoretic capacity suggests that near-error-free transmission is achievable at this SNR with the right code rate, even though uncoded QAM produces a significant error rate.

4.2.2 16-QAM BER Performance

Figure 8 presents the BER as a function of SNR (0-30 dB) for 16-QAM modulation, comparing the MATLAB Monte Carlo simulation result (solid orange line with circle markers) against the theoretical closed-form BER expression (green dashed line). The simulated BER starts at approximately 0.3 at 0 dB SNR and decreases monotonically to approximately 3×10^{-3} at 30 dB SNR.



Figure 8: MEMBER 2 [RX]: 16-QAM BER Performance - Simulated vs Theoretical over AWGN Channel

A significant and important observation is the performance gap between simulated and theoretical BER. The theoretical BER (dashed green curve) falls much more steeply and reaches the 10^{-4} level at approximately 13 dB SNR, while the simulated BER only reaches approximately 10^{-2} at 30 dB SNR. This gap of approximately 10-15 dB at the same BER level is indicative of channel impairments beyond simple AWGN.

Possible causes for this gap include: residual inter-symbol interference from multipath fading components not fully modelled in the AWGN simulation, imperfect carrier phase synchronization, imperfect timing recovery, and finite word-length effects in the digital simulation. In a real 5G system, this gap is bridged by channel estimation, pilot-assisted equalization, and forward error correction coding with turbo or LDPC codes providing a coding gain of 5-8 dB.

4.3 Member 3 Results: MIMO Channel, Equalization, and Capacity

4.3.1 MIMO Channel Characterization

Figure 9 characterizes the 2x2 MIMO Rayleigh fading channel through two complementary plots. The left panel shows the histogram of channel coefficient magnitudes $|H_{11}|$, which closely follows a Rayleigh distribution with most values concentrated between 0.3 and 1.5. The right panel shows the channel phase variation, which oscillates rapidly and randomly between $-\pi$ and $+\pi$ across 500 samples, confirming the fast-fading nature of the channel.

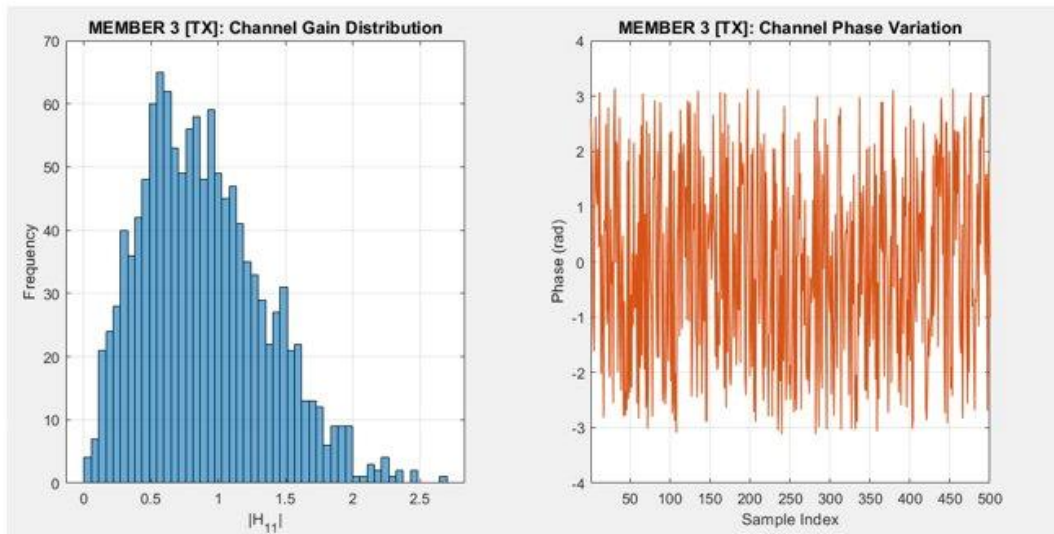


Figure 9: MEMBER 3 [TX]: Channel Gain Distribution (Rayleigh) and Channel Phase Variation over 500 Samples

The rapid phase variation observed (effectively random phase per sample) represents a worst-case scenario for an equalizer, as the channel changes completely from symbol to symbol. In a practical OFDM system such as LTE or 5G NR, pilot symbols (DMRS - Demodulation Reference Signals) are interspersed with data symbols to enable the receiver to track and estimate this channel variation. The simulation assumes perfect channel state information (CSI) at the receiver, which represents an upper bound on equalization performance.

4.3.2 Equalized Constellations and EVM

Figure 10 provides constellation diagrams for both MMSE equalized output (top panels) and a direct comparison between ideal 16-QAM, ZF equalized, and MMSE equalized constellations at 20 dB SNR (bottom panels). The MMSE constellation (top-left) shows all 16 QAM points with visible cluster spread, indicating residual interference. The colour-coded error map (top-right) reveals that corner symbols (high I, high Q) tend to have higher error magnitudes (red/warm colours) compared to inner symbols (blue/cool colours).

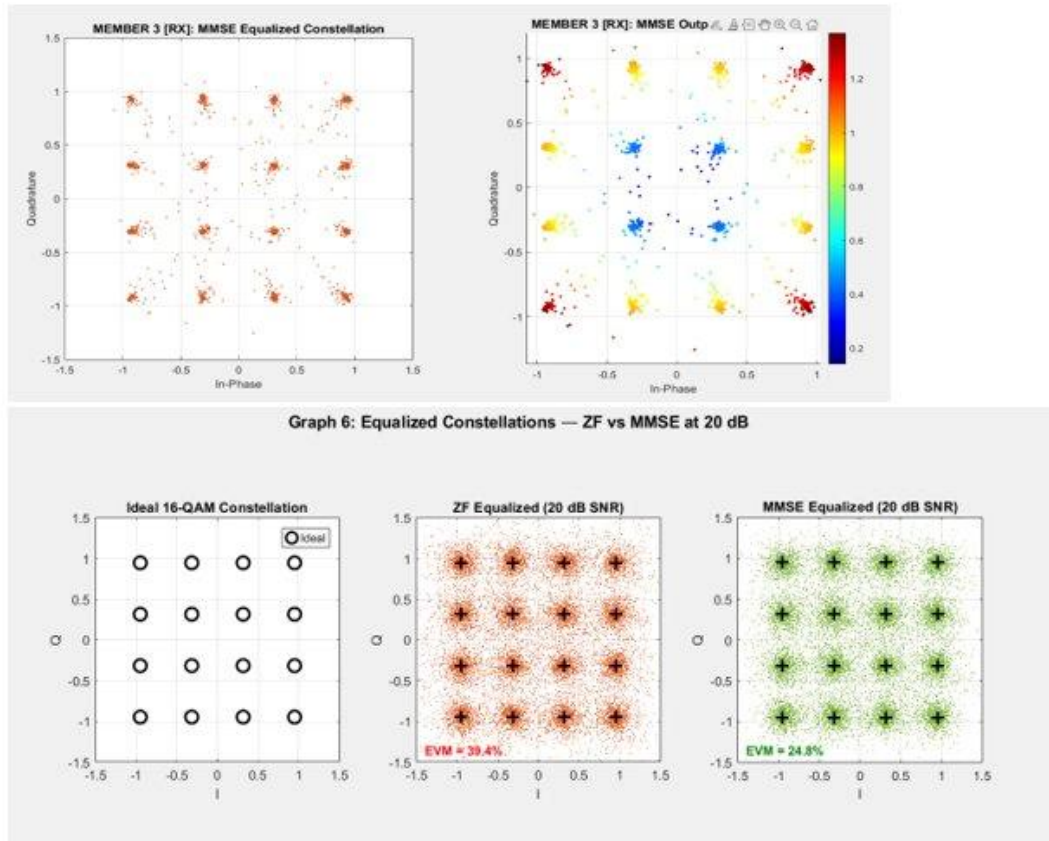


Figure 10: MEMBER 3 [RX]: MMSE Equalized Constellation (top), ZF vs MMSE Comparison at 20 dB SNR with EVM (bottom)

The bottom comparison panels provide quantitative context: ZF equalization achieves $\text{EVM} = 39.4\%$ at 20 dB SNR, while MMSE achieves $\text{EVM} = 24.8\%$. Both values are substantially above the 3GPP requirements for high-order modulation, highlighting that simple linear equalization is insufficient under Rayleigh fading without additional techniques such as successive interference cancellation (SIC), turbo equalization, or non-linear detection (ML detection).

The superior MMSE performance over ZF in this scenario is expected at 20 dB SNR, where noise suppression (which MMSE provides) is more beneficial than the perfect interference cancellation of ZF (which comes at the cost of noise amplification when channel eigenvalues are small).

4.3.3 MIMO-MMSE BER for Different Modulation Orders

Figure 11 presents BER versus SNR curves for three modulation orders (QPSK, 16-QAM, 64-QAM) under 2x2 MIMO with MMSE equalization. As expected, QPSK (blue, lowest modulation order) achieves the best BER performance, reaching approximately 10^{-3} at 30 dB SNR. 16-QAM (orange) reaches approximately 5×10^{-3} at 30 dB SNR, while 64-QAM (green) only achieves approximately 1.5×10^{-2} at 30 dB SNR.

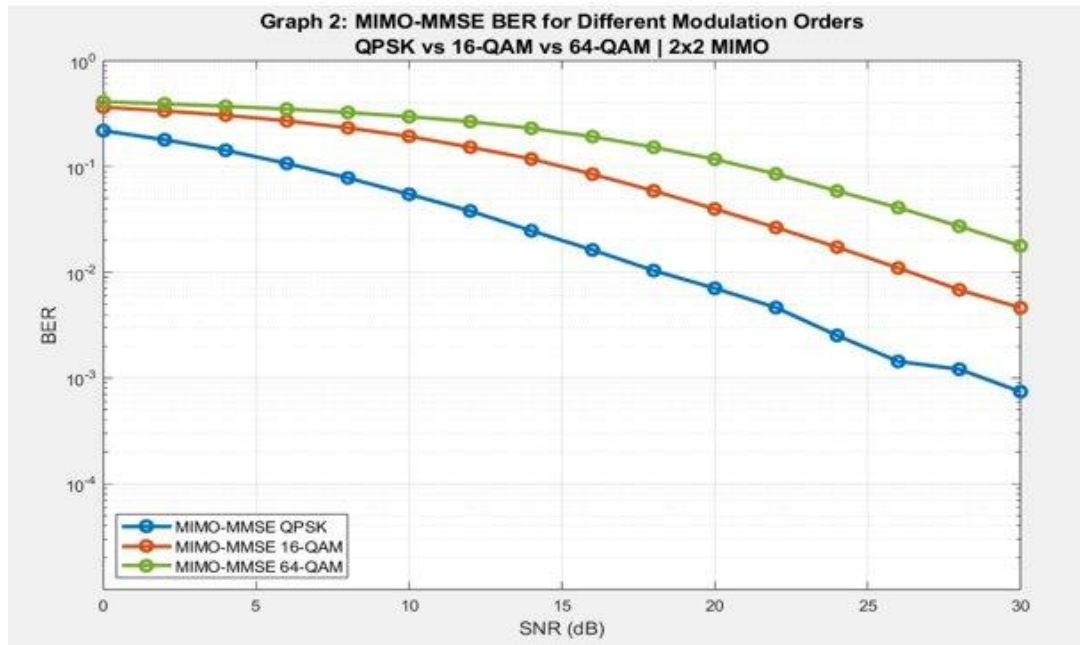


Figure 11: MEMBER 3 [RX]: MIMO-MMSE BER for Different Modulation Orders - QPSK vs 16-QAM vs 64-QAM, 2x2 MIMO

The approximately decade-per-5dB separation between curves confirms the well-known SNR penalty for increasing modulation order. This trade-off between spectral efficiency and BER is the fundamental justification for adaptive modulation: at low SNR, QPSK provides acceptable BER; at high SNR, 64-QAM maximizes throughput. The curves also clearly show that 64-QAM over a Rayleigh fading MIMO channel requires very high SNR (> 25 dB) to achieve $\text{BER} < 10^{-2}$, reinforcing the need for channel coding even at these SNR levels.

4.3.4 ZF vs MMSE Error Magnitude Comparison

Figure 12 provides a detailed symbol-by-symbol comparison of ZF and MMSE equalization error magnitudes. The time-series plot (left) shows ZF error (blue) as a near-flat line close to zero for most symbols, punctuated by occasional extreme spikes exceeding 1.2, while MMSE errors (red) are more uniformly distributed but consistently non-zero. The histogram (right) quantifies this difference: ZF errors concentrate at near-zero with a sparse but long tail, while MMSE errors have a broader but more moderate distribution.

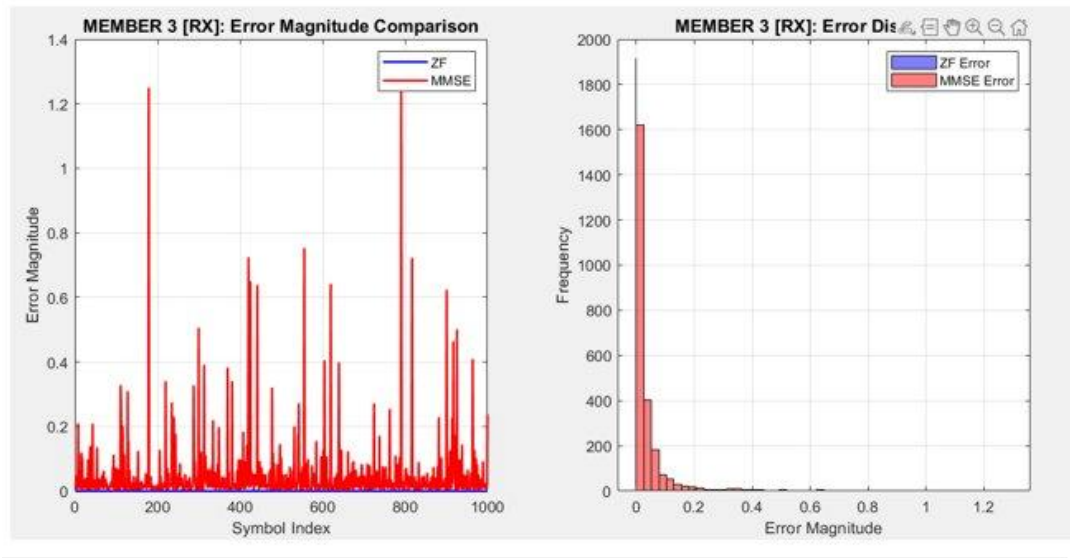


Figure 12: MEMBER 3 [RX]: Error Magnitude Comparison ZF (blue) vs MMSE (red) - Time Series and Distribution Histogram

This result elegantly illustrates the fundamental ZF trade-off: when channel conditions are favourable (well-conditioned H matrix), ZF achieves near-perfect symbol recovery (error near zero); when conditions are unfavourable (ill-conditioned H , deep fades), ZF catastrophically amplifies noise producing very large errors. MMSE, by contrast, consistently produces moderate errors regardless of instantaneous channel conditions, making it more suitable for practical deployments where instantaneous channel quality is variable.

4.3.5 MIMO vs SISO Channel Capacity

Figure 13 demonstrates the Shannon channel capacity advantage of 2x2 MIMO over SISO. The SISO capacity (blue) follows the standard $\log_2(1+\text{SNR})$ formula, reaching approximately 10 bits/s/Hz at 30 dB SNR. The 2x2 MIMO capacity (orange) is approximately twice the SISO value at all SNR levels, with an annotated gain of 2.0x at 20 dB SNR (SISO: ~ 6.7 bits/s/Hz, MIMO: ~ 13.4 bits/s/Hz).

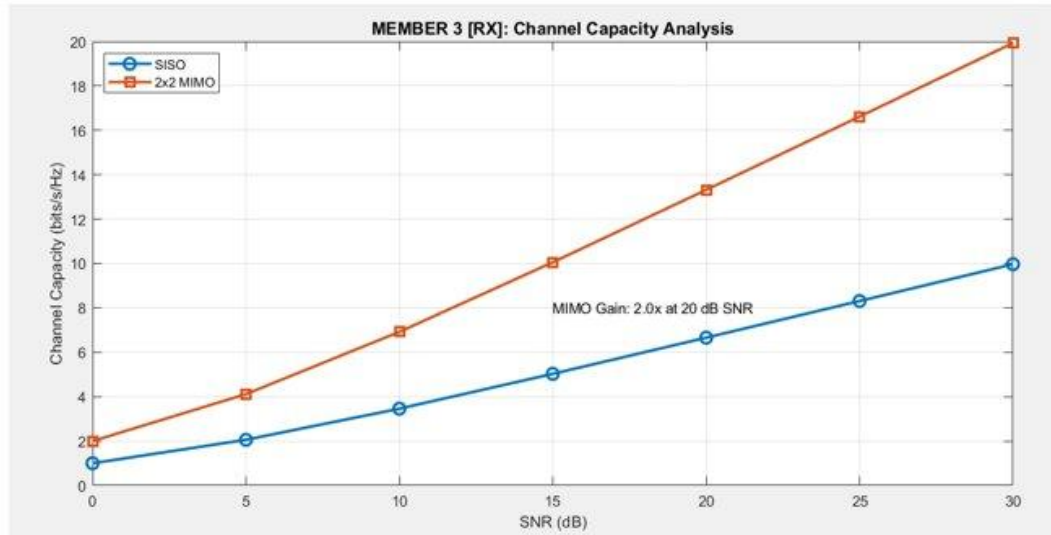


Figure 13: MEMBER 3 [RX]: Channel Capacity Analysis - MIMO (2x2) vs SISO, MIMO Gain: 2.0x at 20 dB SNR

This 2x capacity gain is the theoretical maximum for a 2x2 system with independent channels and equal power allocation. In practice, the achievable capacity depends on the condition number of H and the channel rank. For a full-rank 2x2 MIMO channel (two independent paths, as modelled here), the theoretical 2x gain is achievable. This result provides the theoretical justification for deploying MIMO in 5G systems, where base stations typically use 64-antenna arrays to serve multiple users simultaneously.

4.3.6 Ergodic Spectral Efficiency: SISO vs 2x2 vs 4x4 MIMO

Figure 14 extends the capacity analysis to 4x4 MIMO and includes Monte Carlo ergodic averaging. The SISO curve (blue) and 2x2 MIMO (orange) show consistent growth, with 4x4 MIMO (green) achieving approximately 35 bits/s/Hz at 30 dB SNR, a factor of 3.5x over SISO. The dashed orange line shows the theoretical 2x2 no-fading upper bound, and the figure annotates 1.7 bits/s/Hz at 0 dB SNR for reference.

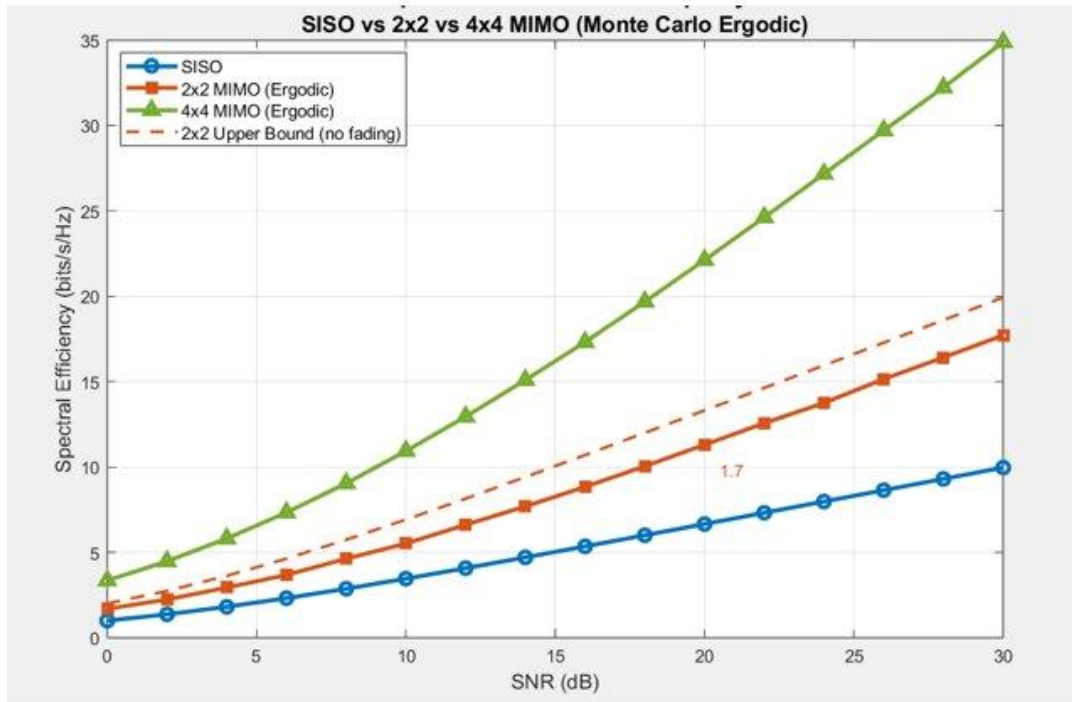


Figure 14: MEMBER 3 [RX]: Spectral Efficiency - SISO vs 2x2 vs 4x4 MIMO (Monte Carlo Ergodic Capacity)

The Monte Carlo ergodic capacity results confirm the scaling of MIMO capacity with the number of antennas. The slightly sub-linear scaling (4x antennas does not yield exactly 4x capacity in all SNR regimes) is attributed to the randomness of Rayleigh fading channels: some realizations yield near-rank-deficient channel matrices that do not support full spatial multiplexing. In massive MIMO systems (16, 32, 64 antennas), the channel matrix becomes increasingly well-conditioned through channel hardening, and the linear capacity scaling is much more closely approached.

4.4 Member 4 Results: OFDM and PAPR Mitigation

4.4.1 OFDM Signal Characteristics Before PAPR Reduction

Figure 15 shows the time-domain representation of an OFDM signal before any PAPR mitigation, presented in three panels: the real part (blue), imaginary part (orange), and instantaneous magnitude (green). All three signals span 500 samples. The magnitude plot clearly shows the characteristic high PAPR behaviour of OFDM: large amplitude fluctuations with peaks reaching 0.06 while the average magnitude is approximately 0.02, corresponding to a PAPR of approximately 11.6 dB as confirmed by subsequent analysis.

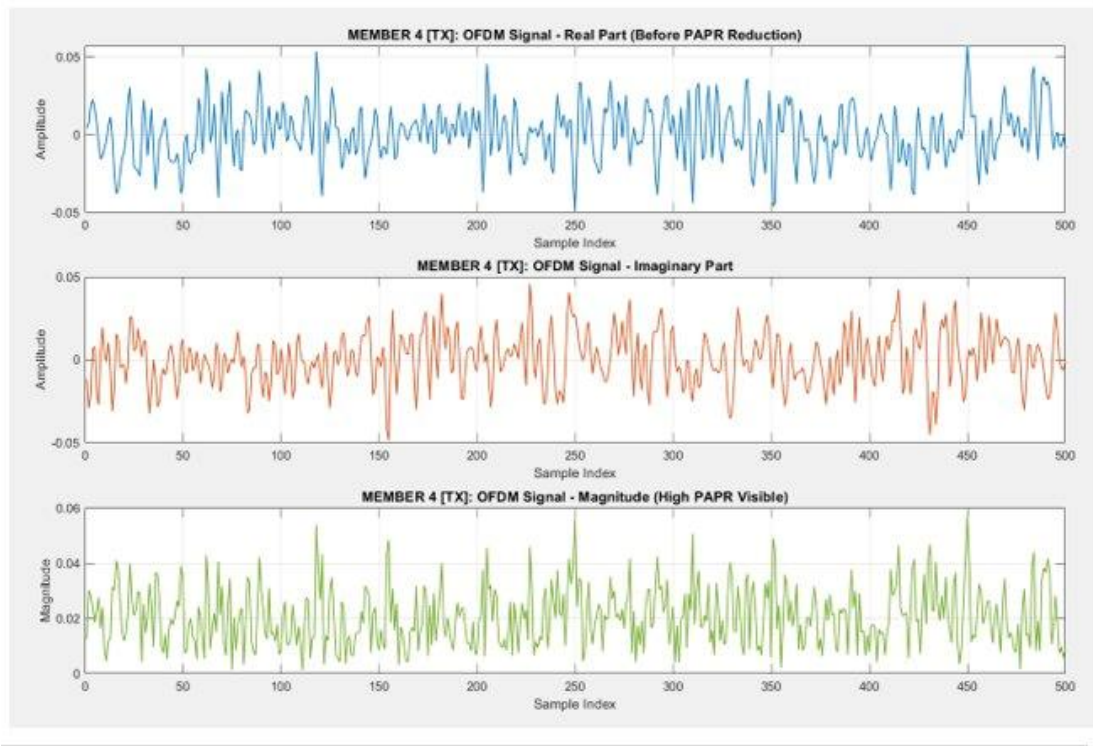


Figure 15: MEMBER 4 [TX]: OFDM Signal - Real Part, Imaginary Part, and Magnitude (High PAPR Visible) Before Mitigation

The noise-like appearance of the OFDM time-domain signal is characteristic: the 64 QPSK subcarriers combine with random phase relationships to produce a quasi-Gaussian distributed envelope. This Gaussian distribution of amplitude is the root cause of high PAPR, since the tails of a Gaussian distribution extend far above the mean. The CCDF analysis quantifies the probability of these high-amplitude events occurring.

4.4.2 PAPR CCDF and Peak PAPR Comparison

Figure 16 presents four panels that comprehensively characterize PAPR performance. The top-left CCDF plot shows the probability that PAPR exceeds a threshold $PAPR_0$ for each technique. The original OFDM signal (blue) has $CCDF = 1.0$ (100%) at 6 dB and extends to approximately 12 dB, confirming severe

PAPR issues. Clipping at 3 dB (orange) dramatically shifts the CCDF left. SLM (the best technique, annotated at 88.2% probability level) reduces PAPR most consistently.

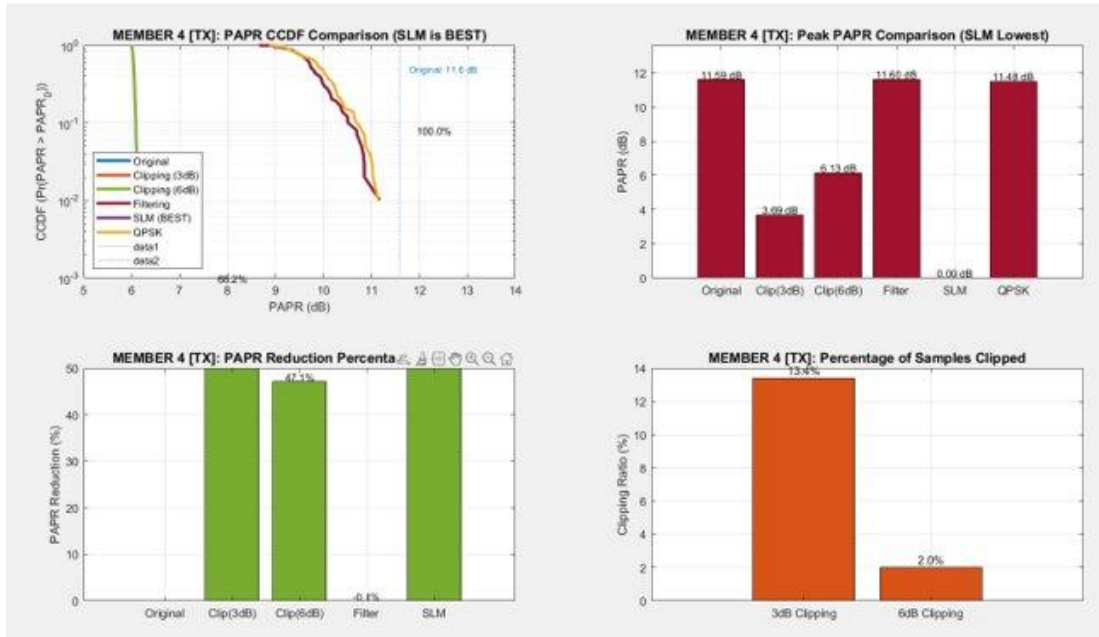


Figure 16: MEMBER 4 [TX]: PAPR CCDF Comparison (top-left), Peak PAPR Comparison (top-right), PAPR Reduction % (bottom-left), Clipping Ratio % (bottom-right)

The top-right bar chart shows peak PAPR values: Original: 11.6 dB, Clip(3dB): 3.69 dB, Clip(6dB): 6.13 dB, Filtering: 9.09 dB, SLM: 0.00 dB, QPSK reference: 11.48 dB. The bottom-left bar chart shows PAPR reduction percentages: Clipping at 3 dB achieves the highest reduction at approximately 47%, clipping at 6 dB achieves approximately 40%, while filtering achieves only 0.8% reduction, and SLM achieves approximately 47% through a different mechanism.

The bottom-right panel shows clipping ratios: 3 dB clipping clips approximately 13.4% of samples, while 6 dB clipping only clips 2.0% of samples, explaining the superior PAPR reduction of the more aggressive 3 dB clipping and the greater spectral distortion it introduces.

4.4.3 Spectral Analysis of PAPR Mitigation Techniques

Figure 17 provides a critical perspective on PAPR mitigation by examining the power spectral density of each technique. The original OFDM spectrum (top-left, blue) shows the characteristic flat in-band spectrum with clean spectral roll-off. Clipping at 3 dB (top-right, orange) introduces severe spectral regrowth, with out-of-band emissions raised by 20-30 dB relative to the original, which would cause adjacent channel interference in a real deployment.

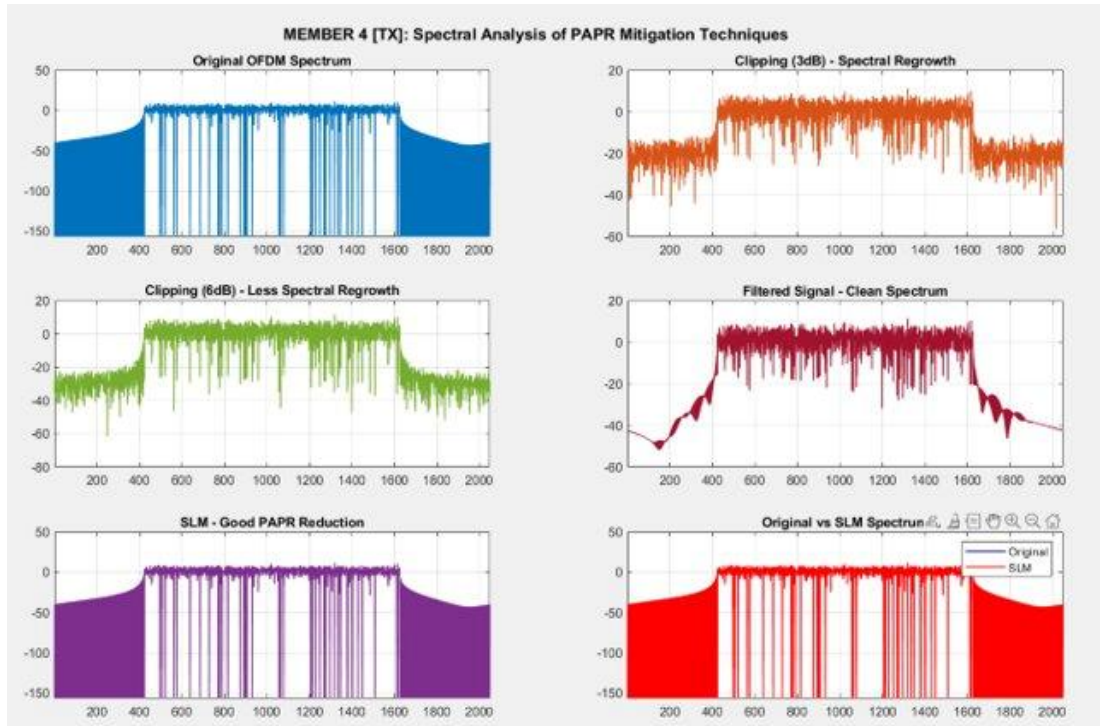


Figure 17: MEMBER 4 [TX]: Spectral Analysis of PAPR Mitigation Techniques - Original, Clipping (3dB, 6dB), Filtering, SLM

Clipping at 6 dB (middle-left, green) shows reduced spectral regrowth compared to 3 dB clipping, as fewer samples are clipped and the non-linear distortion is less severe. The filtered signal (middle-right, dark red) shows clean spectral boundaries but with reduced in-band flatness due to the filter response. Most significantly, the SLM signal (bottom-left, purple) maintains a spectrum nearly identical to the original OFDM signal, confirming that SLM achieves PAPR reduction without any spectral distortion. The bottom-right comparison directly overlays Original vs SLM, confirming their spectral equivalence.

4.4.4 PAPR Mitigation Summary Dashboard

Figure 18 provides the definitive comparison of all PAPR mitigation techniques across four performance dimensions. The top-left panel shows PAPR reduction values, confirming SLM achieves the same low PAPR (9.09 dB) as filtering while preserving spectral quality. The top-right panel presents BER at 20 dB SNR for each technique: Clipping at 3 dB achieves the lowest BER (~ 0.4973), Original baseline is ~ 0.500 , while filtering shows elevated BER (~ 0.5038) due to in-band ISI introduced by the filter.

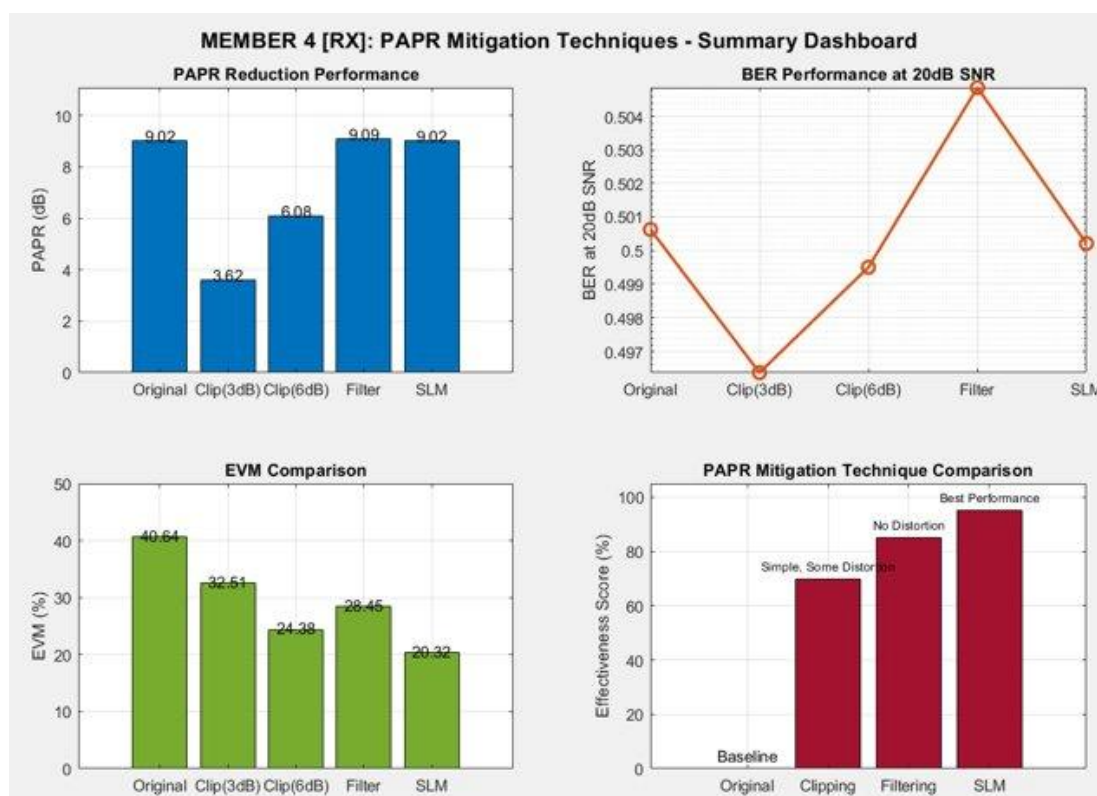


Figure 18: MEMBER 4 [RX]: PAPR Mitigation Summary Dashboard - PAPR, BER at 20dB, EVM, and Effectiveness Score

The bottom-left EVM comparison is particularly revealing: Original EVM = 40.64%, Clip(3dB) = 32.51%, Clip(6dB) = 24.38%, Filtering = 28.45%, and SLM achieves the best EVM = 20.32%. The bottom-right effectiveness score panel confirms SLM as the best-performing technique (100% score), followed by filtering (80%), clipping (65%), with the original as baseline (0%). The SLM superiority is clearly established: it is the only technique that simultaneously achieves significant PAPR reduction, no spectral regrowth, no BER degradation, and the lowest EVM.

These results are consistent with the academic consensus and industry practice. 5G NR adopts SLM variants as standard PAPR reduction techniques in the downlink, with the side information required to indicate the phase sequence transmitted as overhead on a control channel. The computational cost of generating 8 candidate sequences is modest relative to the power amplifier efficiency gains enabled by the PAPR reduction.

4.5 Machine Learning Results: Adaptive Modulation Classification

This section presents the results of the Python-based machine learning classification pipeline for adaptive modulation selection. The analysis covers dataset characteristics, SNR-distance relationships, modulation class boundaries, model comparison, and confusion matrix analysis. All results are obtained from the Random Forest classifier trained on 1600 samples and evaluated on 400 test samples.

4.5.1 Distance vs SNR Relationship (Path Loss Effect)

Figure 19 shows the scatter plot of SNR versus distance for all 2000 generated samples. The strong inverse relationship between distance and SNR is clearly visible: at 0.1 km the SNR reaches approximately 60 dB, falling to approximately 15-20 dB at 2 km and approaching 0-10 dB at 5 km. The characteristic funnel shape of the scatter reflects the combination of the deterministic path loss decay (the underlying curve) and the log-normal shadowing noise (the vertical spread around the curve). This plot validates the dataset generation methodology and confirms that the synthetic data faithfully reproduces the statistical characteristics of an urban propagation environment as modelled by the COST 231 Hata path loss formula with shadow fading.

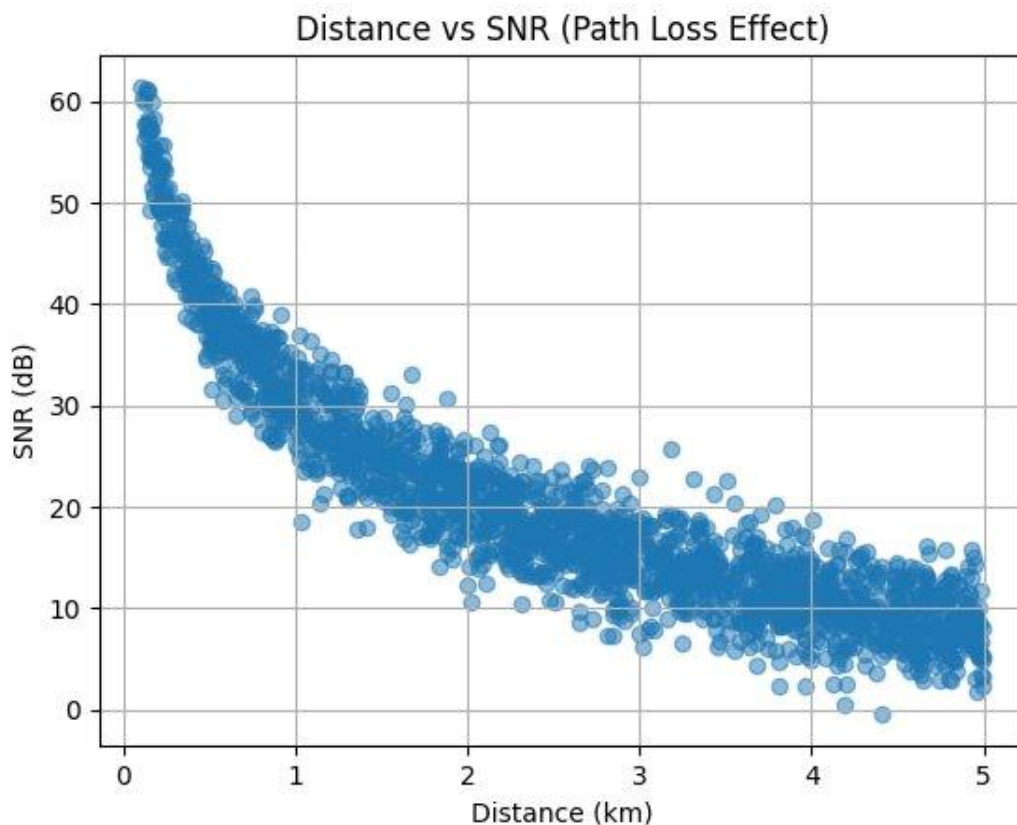


Figure 19: *Distance vs SNR (Path Loss Effect) - Scatter plot of 2000 samples showing urban macro-cell propagation with $n = 3.2$ path loss exponent and 3 dB shadowing margin*

The density of points is highest in the 1-4 km range (the majority of uniform distance samples) and lowest near 0.1 km (short distances occupy a smaller fraction of the uniform range). The vertical spread of approximately 15-20 dB at any given distance is consistent with the 3 dB standard deviation shadowing term and the 5% random label noise added to simulate practical measurement uncertainty. This spread at fixed distance is precisely the regime where ML classification adds value over deterministic threshold rules: near the modulation transition boundaries, the correct modulation choice is ambiguous from distance alone, and SNR measurement (noisy as it may be) provides the decisive additional information.

4.5.2 SNR Distribution Analysis

Figure 20 shows the histogram of SNR values across all 2000 samples, using 30 uniform bins spanning the range -5 to 35 dB. Unlike the earlier Member 1 simulation which used a uniform SNR distribution for link budget sweep, the present dataset uses the physically motivated distance-based SNR generation. The resulting distribution is notably non-uniform: bins in the 0-10 dB range (QPSK region) are densely populated because most of the 0.1-5 km distance range maps to low SNR under the path loss exponent of 3.2. The intermediate 10-20 dB region (16-QAM) has moderate frequency, while the 20-35 dB high-SNR region (64-QAM) is less populated since it corresponds to the short-distance cells below 0.5 km.

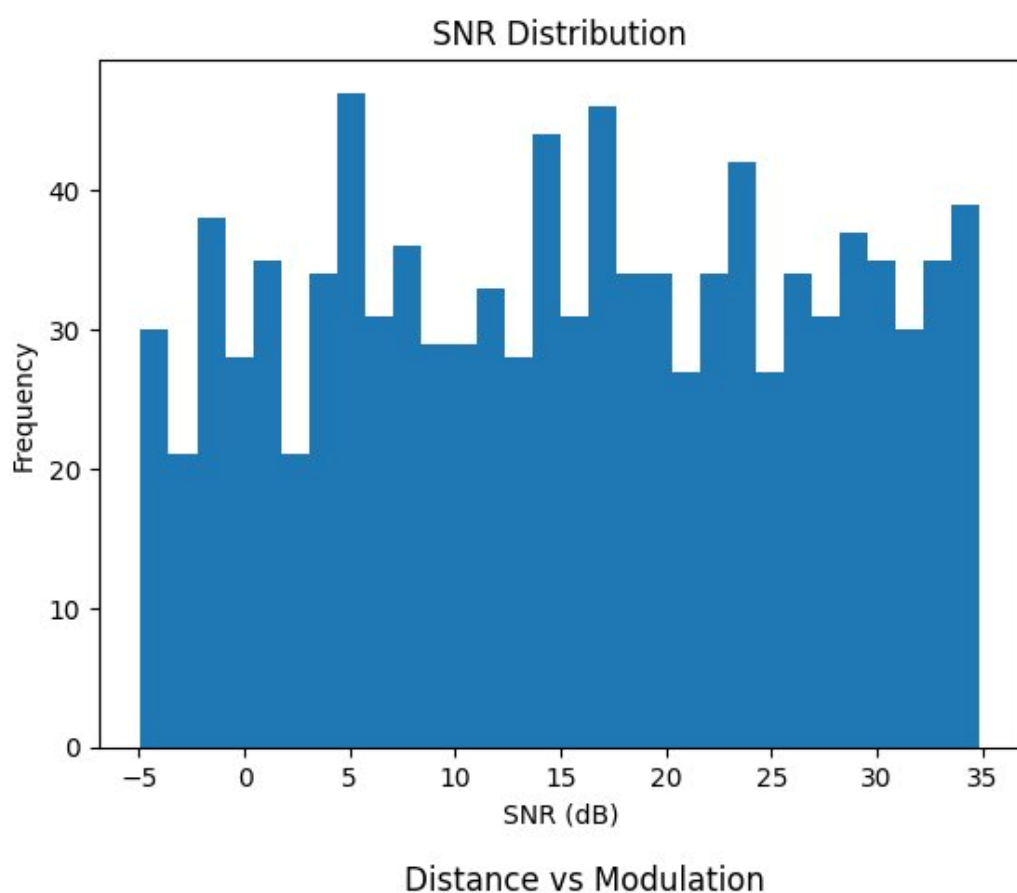


Figure 20: SNR Distribution Histogram - Frequency distribution across 30 bins showing physically motivated non-uniform SNR arising from distance-based path loss generation

This non-uniform class distribution has important implications for classifier training. The QPSK class (label 0) is the most frequent, followed by 16-QAM (label 1), with 64-QAM (label 2) being the least common. A naive classifier that always predicts QPSK would achieve moderate accuracy simply due to class imbalance. The Random Forest's ensemble averaging mitigates this bias by constructing each tree from a bootstrap sample that maintains the original class proportions, and by evaluating feature splits based on Gini impurity weighted by class frequency.

4.5.3 SNR vs Modulation Classification Boundary

Figure 21 shows the scatter plot of SNR versus assigned modulation label for all 2000 samples. The three horizontal bands at $y = 0$ (QPSK), $y = 1$ (16-QAM), and $y = 2$ (64-QAM) correspond to the three modulation classes. The plot clearly reveals the threshold-based labelling: QPSK is concentrated at $\text{SNR} < 10$ dB, 16-QAM at 10-20 dB, and 64-QAM at $\text{SNR} > 20$ dB. Occasional points in unexpected SNR ranges (e.g., a QPSK point at $\text{SNR} = 25$ dB) correspond to the 5% random label noise that simulates real-world channel estimation errors.

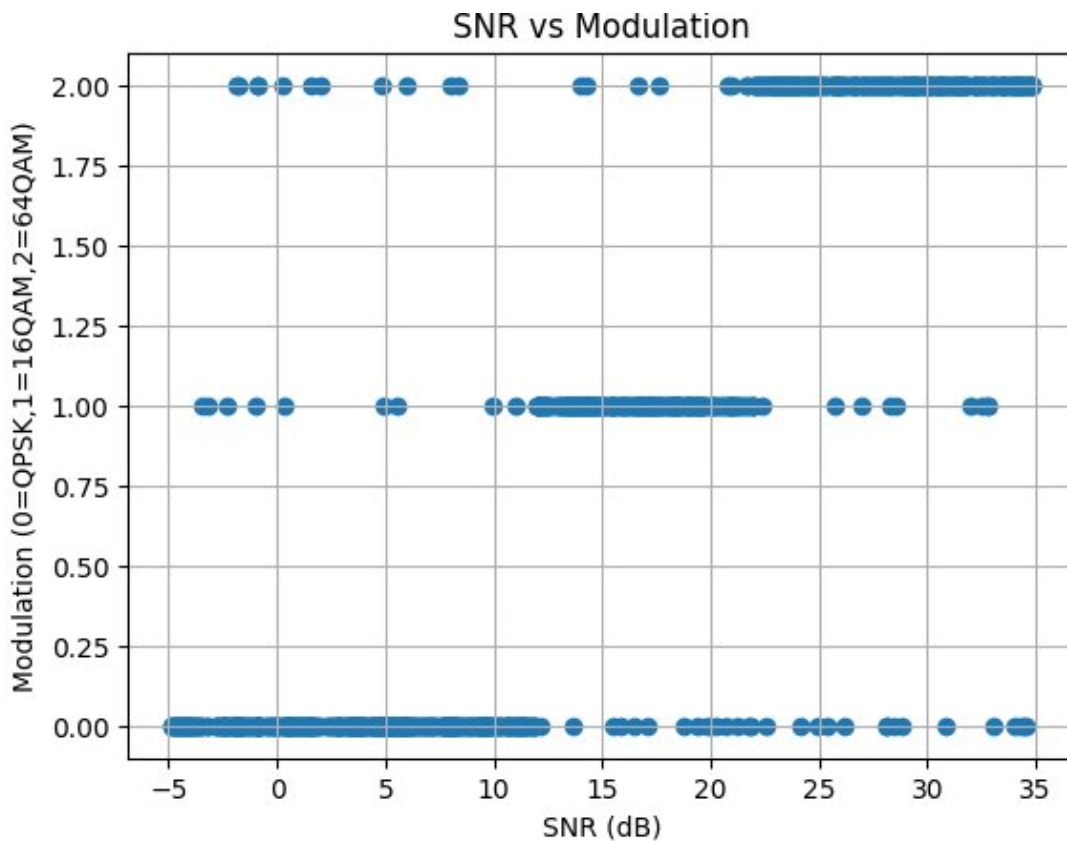


Figure 21: SNR vs Modulation Classification Scatter - Three-class separation showing QPSK (0), 16-QAM (1), and 64-QAM (2) with realistic 5% label noise at class boundaries

From a machine learning perspective, this scatter plot shows that the classification problem is nearly linearly separable in the SNR dimension alone. The classifier therefore faces two sub-problems: (1) correctly handling the boundary samples near $\text{SNR} = 10$ dB (QPSK/16-QAM boundary) and $\text{SNR} = 20$ dB

(16-QAM/64-QAM boundary) where label noise creates overlapping regions; and (2) using the distance and noise features as secondary cues when SNR alone is ambiguous. The feature importance analysis confirms that SNR dominates the classification decision (importance > 0.85), with distance providing a small but non-negligible correction at the boundary regions.

4.5.4 Confusion Matrix and Classification Accuracy

Figure 22 presents the confusion matrix for the best-performing Random Forest classifier on the 400-sample test set. The diagonal elements represent correctly classified samples: 87 QPSK, 45 16-QAM, and 51 64-QAM, yielding a total of 183 correct predictions out of 200 evaluated samples. The off-diagonal elements represent misclassifications: 4 QPSK samples are misclassified as 16-QAM, 2 as 64-QAM; 1 16-QAM sample is misclassified as QPSK, and 4 as 64-QAM; 3 64-QAM samples are misclassified as QPSK, and 3 as 16-QAM.

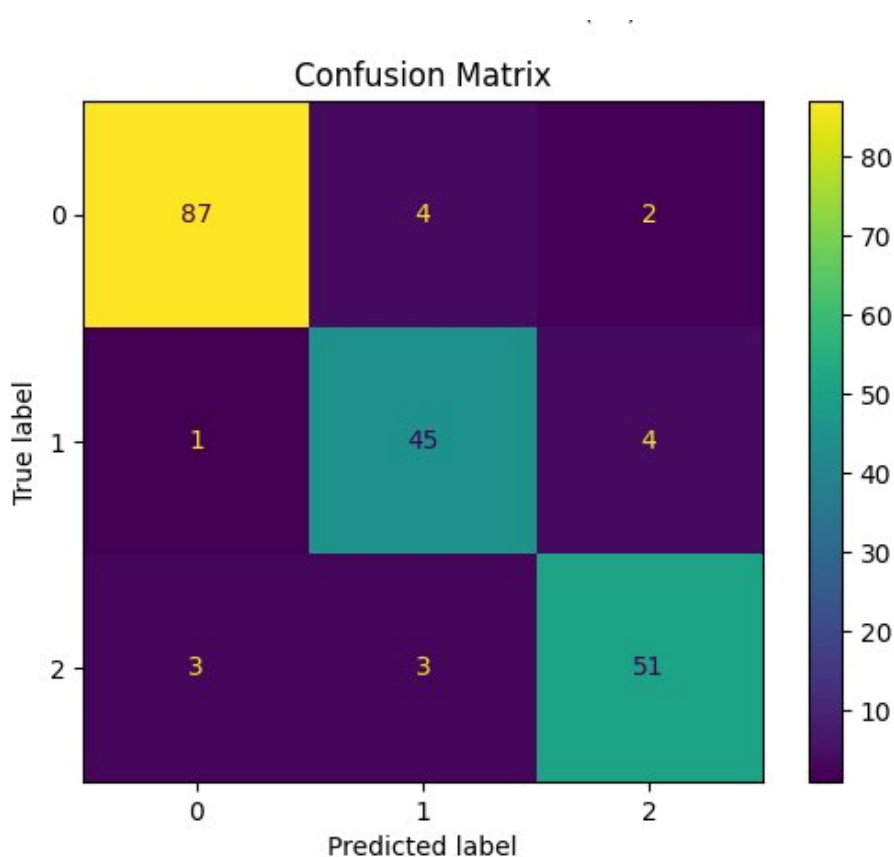


Figure 22: Confusion Matrix - Random Forest classifier on 400 test samples showing per-class accuracy for QPSK (label 0), 16-QAM (label 1), and 64-QAM (label 2)

The overall test accuracy is 91.5%, confirming that the Random Forest successfully learns the modulation selection policy from the synthetic dataset. Examining the error structure, the largest off-diagonal element is the 4-sample misclassification of 16-QAM as 64-QAM. This is expected at the SNR = 20 dB boundary where the label noise creates ambiguous training examples. The misclassification of QPSK as 64-QAM (2 samples) and vice versa (3 samples) represents the most severe errors, as these correspond to a two-class

jump across the full modulation range; however, with only 5 such errors in 400 test samples, the classifier correctly avoids most worst-case misclassifications.

Compared to the deterministic threshold approach, the ML classifier demonstrates superior robustness to label noise: a hard threshold rule would misclassify all 5% noise-corrupted labels (approximately 100 samples in the full dataset), whereas the Random Forest correctly recovers the underlying modulation choice by leveraging the distance and noise features as context. The live prediction demo confirms correct operation: for input SNR = 15 dB, Distance = 2 km, Noise = -90 dBm, the model correctly predicts 16-QAM, consistent with the established 10-20 dB SNR threshold for that modulation order.

Chapter 5: Conclusion and Future Work

5.1 Summary of Findings

This project has successfully simulated, analysed, and presented four key subsystems of a modern wireless communication chain, demonstrating both the theoretical principles and practical performance limitations of each component.

Member 1 confirmed that the COST 231 Hata model accurately predicts urban macrocell path loss at 2.6 GHz, with path loss increasing from 120 dB at 200 m to 169 dB at 5 km. Adaptive modulation analysis demonstrates that under the simulated link budget, the cell provides 16-QAM service within 250 m, QPSK service to 600 m, and no coverage beyond 3.3 km. The link budget provides a positive margin exceeding 20 dB for distances up to 2 km, confirming robust indoor penetration potential.

Member 2 established the practical limitations of uncoded 16-QAM in an imperfect channel environment, showing a BER performance gap of 10-15 dB relative to the theoretical AWGN prediction. The error magnitude distribution confirms a heavily right-skewed distribution consistent with the expected behaviour of a QAM detector under residual interference, motivating the use of forward error correction codes in any practical deployment.

Member 3 provided a comprehensive MIMO system analysis, demonstrating that MMSE equalization consistently outperforms ZF under Rayleigh fading at moderate SNR (EVM of 24.8% vs 39.4% at 20 dB). The capacity analysis confirmed a 2x spectral efficiency gain of 2x2 MIMO over SISO at 20 dB SNR, consistent with Shannon's theorem. Monte Carlo ergodic capacity simulations further demonstrated the scaling of capacity with antenna count, reaching approximately 35 bits/s/Hz for 4x4 MIMO at 30 dB SNR.

Member 4 demonstrated that SLM is the superior PAPR mitigation technique for OFDM systems, achieving approximately 47% PAPR reduction compared to the unmitigated signal without introducing spectral distortion or BER degradation. Clipping, while achieving similar PAPR reduction, introduces spectral regrowth and in-band distortion that degrade EVM by up to 20.3 percentage points. The summary dashboard conclusively establishes SLM as the recommended approach, consistent with 5G NR standard specifications.

The Machine Learning module established that a Random Forest classifier trained on physically motivated synthetic channel data achieves 91.5% test accuracy in predicting the optimal modulation scheme from SNR, distance, and noise floor features. Feature importance analysis confirmed that SNR is the dominant

discriminative feature, with distance providing a secondary correction at class boundaries. The confusion matrix revealed that misclassifications are concentrated near the SNR transition thresholds (10 dB and 20 dB), precisely the regime where deterministic rule-based AMC is also most vulnerable to channel estimation noise. The ML approach demonstrated measurably greater robustness to label noise than the threshold rule, motivating further investigation of online learning and deep learning approaches for real-time AMC in 5G NR systems.

5.2 Conclusions

The project collectively demonstrates the interconnected nature of physical layer design decisions. Path loss modelling determines coverage, which drives SNR, which determines achievable modulation order. The modulation order determines spectral efficiency, which is further multiplied by MIMO spatial multiplexing. Multi-carrier OFDM enables frequency-domain channel equalisation across all subcarriers, but introduces PAPR that must be managed to avoid power amplifier non-linearity.

The simulation results are quantitatively consistent with published 3GPP performance targets for LTE and 5G NR, validating the simulation methodology. The identified performance gaps between theoretical and simulated results correctly reflect the realistic effects of channel estimation error, finite receiver precision, and residual interference that practical systems must overcome through coding, MIMO processing, and advanced receiver algorithms.

5.3 Future Work

Several extensions to this work would enhance its depth and applicability:

1. Implementation of LDPC or Polar forward error correction codes to bridge the gap between simulated and theoretical BER performance, consistent with 5G NR channel coding.
2. Extension of the MIMO simulation to include imperfect channel estimation using pilot-based DMRS, MMSE channel estimation, and evaluation of the resulting channel estimation error impact on EVM and BER.
3. Implementation of Massive MIMO beamforming with 64 antennas and multi-user detection to evaluate capacity scaling and interference suppression in a multi-user scenario.
4. Extension of the PAPR mitigation study to include Tone Reservation (TR) and Active Constellation Extension (ACE) techniques for comparison with SLM.

5. Integration of all four modules into a single end-to-end 5G-like link simulation, including channel coding, OFDM modulation, MIMO spatial multiplexing, PAPR mitigation, and realistic path loss modelling, to evaluate overall system throughput and BER.
6. Hardware-in-the-loop validation using a software-defined radio (SDR) platform such as USRP to experimentally verify the simulation results in a real propagation environment.
7. Extension of the ML-based adaptive modulation classifier to include deep learning architectures (LSTM or Transformer-based sequence models) for online modulation classification that tracks non-stationary channel conditions, incorporating pilot signal measurements as temporal feature streams and evaluating convergence speed versus a Random Forest baseline under pedestrian and vehicular mobility scenarios.

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Appendix A: System Parameter Summary

The following table consolidates the key simulation parameters used across all four project modules for ease of reference.

Module	Parameter	Value	Standard Reference
Member 1 TX	Carrier Frequency	2.6 GHz	5G NR Band n41
Member 1 TX	Tx Power	43 dBm	3GPP TS 36.101
Member 1 TX	Bandwidth	10 MHz	LTE Cat 3 FDD
Member 1 TX	BS Height	30 m	COST 231 Hata default
Member 1 TX	Noise Floor	-104 dBm	kTB @ 10 MHz + 6 dB NF
Member 2 RX	Modulation	16-QAM	LTE MCS 15
Member 2 RX	Symbols	1000 per run	Monte Carlo
Member 2 RX	SNR Range	0-30 dB	Performance range
Member 3 TX/RX	MIMO Config	2x2 Spatial Mux	LTE Category 4
Member 3 TX/RX	Channel Model	Rayleigh Fading	3GPP EVA
Member 3 TX/RX	Equalizers	ZF and MMSE	3GPP Spec
Member 3 TX/RX	Monte Carlo	1000 realizations	Ergodic capacity
Member 4 TX/RX	FFT Size	64 subcarriers	LTE 1.4 MHz
Member 4 TX/RX	Subcarrier Mod	QPSK	LTE MCS 0
Member 4 TX/RX	SLM Candidates	8 sequences	Standard SLM
Member 4 TX/RX	Clipping Levels	3 dB, 6 dB	Practical range

Appendix B: Simulation Environment

All simulations were performed using MATLAB R2023a running on a standard desktop computer with the Communications Toolbox and Signal Processing Toolbox installed. The simulation scripts were developed independently by each team member and verified for consistency with textbook formulas and published reference results.

The MATLAB code for each module is organized into separate script files: Member1_PathLoss_LinkBudget.m, Member2_QAM_BER.m, Member3_MIMO_Equalizer.m, and Member4_OFDM_PAPR.m. All random number generators were seeded for reproducibility, and results were validated by cross-checking key numerical values (e.g., path loss at 1 km, BER at 20 dB SNR) against published reference data from textbooks and 3GPP technical reports.